



UNIVERSIDAD DE INVESTIGACIÓN DE TECNOLOGÍA EXPERIMENTAL YACHAY

Escuela de Ciencias Matemáticas y Computacionales

TITULO

Sistema Integrado de Inteligencia de Negocio para la Predicción de
Crisis Crediticias: Deep Learning Multi-Horizonte, Datamart Analítico
y Asistente Conversacional con Superset MCP

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Dedication

To Anita María, Clarisse and Josué.

Wilman Roberto Suárez Zambrano

Acknowledgment

To God, for granting me everything.

To Zenaida, your support has been invaluable.

To Juan Pablo, for your assistance and dedication to this work.

To my family, thank you for your constant support.

To my professors from the Master's program.

To my colleagues at ECMC (...).

To all of you, my deepest gratitude.

Wilman Roberto Suárez Zambrano

Resumen

En escenarios de desastre, la ausencia de infraestructura y el comportamiento dinámico de los nodos móviles convierten la comunicación fiable en un desafío crítico. Este trabajo propone un enfoque basado en inteligencia artificial para mejorar las comunicaciones en redes inalámbricas ad hoc bajo tales condiciones. Se construyó un conjunto de datos desde cero empleando el simulador **ns-3** para el modelado de la capa de comunicación y **BonnMotion** para simular la movilidad humana en situaciones de emergencia. Se extrajeron dos características clave—*factor de utilización del canal* y *tamaño de la cola de paquetes*—que sirvieron para entrenar un modelo supervisado basado en **CatBoost**. Tras validar mediante Accuracy, Precision y F1-Score, el modelo se integró de nuevo en el entorno de simulación. Los resultados experimentales muestran que la estrategia impulsada por IA mejora de forma significativa el **Packet Delivery Ratio (PDR)**, el **throughput** y el **End-to-End Delay** en comparación con un escenario base sin mecanismos de Calidad de Servicio (QoS). Estos hallazgos demuestran la viabilidad de incorporar decisiones inteligentes en las capas de comunicación para aplicaciones de respuesta ante desastres y ponen de relieve su robustez frente a patrones de movilidad diversos.

Palabras Clave:

Mobile Ad hoc Networks; Disaster communications; Machine Learning; CatBoost; Packet Delivery Ratio; ns-3 simulation; Quality of service;

Abstract

In disaster scenarios, reliable communication becomes a critical challenge due to the absence of infrastructure and the dynamic behavior of mobile nodes. This work proposes an artificial intelligence-based approach to enhance communication in wireless ad hoc networks under such conditions. We constructed a dataset from scratch using the **ns-3** simulator for communication modeling and **BonnMotion** for simulating human mobility in disaster situations. Two key features—*channel utilization factor* and *queue packet size*—were extracted and used to train a supervised learning model based on the **CatBoost** algorithm. After validation through accuracy, precision, and F1-score, the model was exported back into the simulation environment. Experimental results show that the AI-enhanced strategy significantly improves **Packet Delivery Ratio (PDR)**, **Throughput**, and **End-to-End Delay** compared with a baseline scenario lacking Quality-of-Service (QoS) mechanisms. These findings demonstrate the feasibility of integrating intelligent decision-making into communication layers for disaster-response applications and underscore its robustness across diverse mobility patterns.

Keywords:

Mobile Ad hoc Networks; Disaster communications; Machine Learning; CatBoost; Packet Delivery Ratio; ns-3 simulation; Quality of service;

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Chapter 1

Introduction

1.1 Antecedentes

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1.2 Background

Over the past decade, the frequency and severity of disasters—such as major earthquakes, hurricanes, wildfires, and attacks on critical infrastructure—has risen markedly, disabling conventional telecommunications networks within minutes [1]

[?, ?]. In the absence of this infrastructure, rescue teams require rapid, self-sufficient solutions that allow them to coordinate operations, share aerial imagery, and transmit sensor data. **Mobile Ad Hoc Networks (MANETs)** have emerged as an ideal alternative: each device acts simultaneously as an end node and a router, forming a self-organising mesh that operates without towers or backbone links [?]. Field trials conducted after the 2017 Puebla earthquake showed that a MANET of 60 handheld radios restored basic voice and messaging services in under thirty minutes, although it suffered from latency and packet-loss issues caused by chaotic mobility and signal shadowing from collapsed buildings [?].

In response to these limitations, the scientific community has incorporated **artificial intelligence** (AI) techniques into the protocol stack. Machine-learning models can predict link breakages, balance traffic load, and adjust transmission power in real time [?, ?]. Deep-reinforcement-learning algorithms have outperformed classical routing schemes by anticipating network fragmentation and proactively rerouting data flows [?]. Unsupervised methods—such as *k-means* clustering—group nearby nodes and reduce signalling overhead when the network becomes dense and heterogeneous [?]. However, many of these studies rely on synthetic mobility patterns that overlook the actual dynamics of affected populations, potentially leading to overly optimistic quality-of-service estimates [?].

To close this realism gap, researchers have developed **data-driven mobility models** and high-fidelity simulators. The *Disaster Area Mobility Model* (DA-MM) in BonnMotion reproduces the movement of victims and rescuers through rubble zones, shelters, and command posts [?]. When combined with Carpenter and Sichitiu’s obstacle-shadowing model in **ns-3**, DA-MM corrects packet-delivery overestimation by accounting for the attenuation caused by concrete blocks and collapsed structures [?]. Recent studies have incorporated real evacuation traces—such as those collected during Hurricane Ian in 2022—showing that mobility surges can triple network congestion in less than two hours [?]. Meanwhile, the European guidelines for trustworthy AI underscore the need for transparency, fairness, and robustness—factors that are critical when network decisions can directly affect human safety [?].

In summary, the convergence of **MANETs, realistic simulation, and AI algorithms** points the way to networks capable of sustaining vital communications when conventional means fail. Nonetheless, important challenges remain: training models with data that truly represent critical scenarios, ensuring an ethical and explainable operation, and validating—at least in simulation or laboratory settings—that the proposed solutions reduce latency and packet loss without overloading frontline devices. This study tackles these issues and lays the groundwork for future field trials, once the initial simulation or lab stages have demonstrated their effectiveness.

1.3 Problem statement

Natural and human-made disasters—such as earthquakes, floods, wildfires, or targeted attacks on critical infrastructure—can disable conventional telecommunications networks in a matter of minutes. When this disruption occurs, rescue teams are left without reliable channels to coordinate operations, share aerial imagery, report victim locations, or transmit sensor data that warns of aftershocks and fire hotspots. In the absence of fixed infrastructure, *ad hoc* networks emerge as the immediate alternative: each device acts simultaneously as an end node and a router, forming self-organising meshes that can be deployed within minutes. Nevertheless, rapidly changing topology, radio-frequency interference, and device heterogeneity lead to high latencies and packet losses that undermine the usefulness of the transmitted information.

To mitigate these limitations, two essential challenges must be addressed: **(i)** dynamically and robustly selecting routes in highly mobile and fragmented environments, and **(ii)** adapting spectrum access under adverse conditions of interference and obstacles. Machine-learning techniques offer a promising framework because they can uncover connectivity patterns and anticipate link breakages using historical data and realistic simulations. The aim of this thesis is therefore to implement and validate a *machine-learning* model that, when trained with data representative of disaster scenarios, proves its ability to reduce latency and packet loss without significantly increasing control signalling. In doing so, it seeks to improve the quality of service of *ad hoc* networks precisely when they are needed the most—after conventional infrastructure has failed.

1.4 Objectives

1.4.1 General Objective

Implement machine learning models to optimize data transmission in disaster scenarios with ad-hoc networks. This approach aims to minimize latency and packet loss (while improving throughput), ensuring the efficient transmission of information, especially under disaster conditions.

1.4.2 Specific Objectives

1. Conduct a comprehensive literature review on the current state of wireless communications in disaster scenarios. This review will help identify existing limitations and highlight opportunities where artificial intelligence can enhance network performance.
2. Collect a dataset that characterizes communication behavior in disaster scenarios using mobility and wireless network simulators. This dataset will serve as a foundational input for designing and training machine learning models tailored to emergency environments.
3. Preprocess the collected data to extract suitable input and output features for the machine learning model. This step includes feature selection, data normalization, and organization to ensure optimal learning and model efficiency.
4. Select and train a machine learning model that improves key performance metrics such as *Accuracy* and *F1-Score*. For this purpose, the architecture of a supervised algorithm will be found using hyperparameter optimisation algorithms.
5. Simulate and validate the trained model in realistic disaster scenarios using metrics such as packet delivery ratio and end-to-end delay. These simulations will assess the model's real-world impact on the quality of service (QoS) in ad hoc wireless networks under challenging conditions.

Chapter 2

Theoretical Framework

This chapter lays the conceptual groundwork for the thesis: it first outlines the principles of wireless networks operating in disaster situations, then discusses the mobility models and simulation tools that replicate those environments, and finally reviews recent Machine Learning contributions aimed at improving communication performance in *ad-hoc* networks.

En la figura 2.1 dice esto

Ad-hoc networks are viewed as the fastest way to restore communications when conventional infrastructure is out of service. Their performance, however, depends on factors such as unpredictable human movement, the presence of obstacles, and environmental interference. To capture these conditions realistically, the research community combines disaster-specific mobility models—most notably BonnMotion’s *Disaster Area Mobility Model*—with discrete-event simulators like **ns-3**, to which obstacle-attenuation or shadow-fading models can be coupled [?, ?, ?].

In recent years, **Machine Learning techniques** have been explored to enable the network to adapt autonomously to changing topologies. Several studies show that models trained on simulated or field data help select more stable routes, adjust medium-access parameters, and prioritise critical flows—all without human intervention [?]. These approaches give the network a self-organising character, capable of redistributing resources whenever conditions become adverse, although they still depend on the quality of the training data and the fidelity of the simulated environment [?].



Figure 2.1: Figura de audios

2.1 Mobile Ad Hoc Networks (MANETs)

2.1.1 Definition and General Characteristics

Mobile Ad Hoc Networks (MANETs) are decentralised wireless communication systems composed of mobile nodes that communicate with one another without any fixed infrastructure or central administration. Each node acts simultaneously as a host and a router, enabling data to be forwarded through multiple hops between nodes that are not within direct transmission range. **Figure 2.2** illustrates a typical MANET topology and the multi-hop cooperation among nodes.

These networks are characterised by their ability for **self-organisation**, **self-configuration**, and **self-healing**, which makes them especially useful in dynamic or unpredictable environments where conventional infrastructure deployment is not feasible. Such scenarios include military operations, areas affected by natural disasters, intelligent vehicular networks, and temporary deployments at large-scale events [?].

Thanks to their flexibility and adaptability, MANETs have been widely studied in the academic community and remain an active research area, particularly in the development of efficient and secure routing protocols.

The main characteristics of Mobile Ad Hoc Networks (MANETs) include:



Figure 2.2: Example of a MANET topology [?].

- **Variable-capacity wireless links:** Nodes communicate over wireless links whenever they are within mutual coverage range. These links can exhibit different capacities because nodes may carry multiple radio interfaces operating on various frequency bands and offering distinct transmission and reception capabilities.
- **Mobility and dynamic topology:** Nodes can move freely, causing continuous changes in network connectivity. Some nodes may be highly mobile—for example, a device in a moving vehicle—whereas others may remain relatively static. This results in a dynamic topology in which nodes can join or leave the system at any time.
- **Self-organisation, self-configuration, and autonomy:** The network is created, managed, and adapted dynamically by the nodes themselves, without external intervention or the need for centralised administration or fixed infrastructure. This includes the self-configuration of essential parameters such as addressing, routing, position identification, and power control.
- **No fixed infrastructure or central administrator:** MANETs do not require access points, central routers, or servers to operate. This enables rapid deployment,

particularly in environments where traditional infrastructure is unavailable or has been destroyed.

- **Collaboration among nodes:** Each node functions not only as a sender and receiver but also as a router, forwarding packets that are not addressed to itself. This cooperation is essential for maintaining connectivity and overall network operation in the absence of dedicated routing nodes.
- **Resource and energy constraints:** Nodes in a MANET are typically battery-powered mobile devices with limited processing, storage, and bandwidth capabilities. These restrictions critically influence the design of algorithms and protocols, which must be highly efficient in both energy consumption and resource utilisation.
- **Heterogeneity:** Devices that make up a MANET can be highly diverse—such as smartphones, laptops, tablets, or sensors—each with different hardware capabilities, energy reserves, and mobility patterns.

2.2 Most Common Topologies

MANETs can assume various configurations depending on the placement and mobility of their nodes. Understanding these topologies is crucial for designing efficient routing protocols and control mechanisms. The three most common arrangements are described below and illustrated in Figures 2.3, 2.4, and 2.5.

2.2.1 Linear Topology

The linear arrangement (Figure 2.3) occurs when nodes are aligned—for instance, along an underground corridor, a vehicular tunnel, or a line of rescuers advancing through a narrow hallway. Its main limitation is sequential dependency: the failure of an intermediate node can interrupt communication between the endpoints, increasing latency because alternative routes must be established.



Figure 2.3: Typical linear topology in MANETs.

2.2.2 Mesh Topology

The mesh structure (Figure 2.4) provides multiple redundant paths between pairs of nodes. This increases fault tolerance and facilitates load balancing, albeit at the cost of greater routing complexity. It is the preferred configuration in search-and-rescue deployments or UAV swarms that cover a broad area with uniform density.



Figure 2.4: Mesh topology with redundant links.

2.2.3 Star Topology

In the star arrangement (Figure 2.5), a central node—such as a high-capacity drone or a command vehicle—acts as a hub and concentrates all traffic. This topology simplifies routing and reduces latency toward the centre, but it introduces a single point of failure: if the central node is compromised, the network fragments.



Figure 2.5: Star topology with a central hub node.

2.2.4 Applications in Emergency Contexts

MANETs prove especially advantageous in disaster scenarios—earthquakes, hurricanes, or floods—where conventional infrastructure is out of service. Their self-organising capability enables emergency teams to re-establish basic communications, transmit critical data, and extend coverage within minutes, without relying on backbone links or fixed access points.

Table 2.1 summarises the most representative domains in which ad-hoc networks have been deployed or studied and will serve as a guiding thread for the examples described below.

Ad-hoc networks have become a cornerstone for the **rapid restoration of connectivity** after earthquakes, floods, or complete cellular-infrastructure blackouts. Field-deployment experiments—for instance, the **R-RDSP** system built on off-the-shelf Wi-Fi devices—show that an autonomous mesh can restore basic voice and data services in under ten minutes and reduce end-to-end delay by 14.5% compared with earlier schemes [?]. These plug-and-play architectures remove the need for root nodes and, thanks to multi-hop broadcasting, keep rescue teams online when conventional routes collapse.

Domain	Typical Applications
Military	Tactical communications; autonomous operations
Emergency Response	Search and rescue; temporary emergency networks
Scientific / Environmental	Environmental monitoring; precision agriculture
Education / Entertainment	Virtual classrooms; multi-user gaming
Vehicular	Vehicle-to-vehicle communication; traffic alerts
Commercial	Mobile offices; e-commerce on the move

Table 2.1: Representative application domains of MANETs.

A second cornerstone is **dynamic aerial coverage** using swarms of unmanned vehicles. Chandran and Vipin describe how *multi-UAV FANETs* extend line-of-sight and relay 4 K video over devastated areas, while also discussing formation and security challenges [?]. Complementing their work, Li *et al.* propose a multi-objective deployment algorithm that balances coverage, energy consumption, and path redundancy; their strategy improves the packet-delivery ratio by $\uparrow 11\%$ in ns-3 wild-fire simulations [?]. These aerial solutions serve as *backhaul* nodes for ground sensors, forming fault-tolerant hybrid networks.

Artificial intelligence has further enhanced these applications by optimising routing under highly variable conditions. Jin *et al.* propose *DRLRR*, a resilient routing protocol based on deep reinforcement learning that, in simulated post-earthquake urban scenarios, raises the *Packet Delivery Ratio* by 18 % and cuts end-to-end delay by 22 % compared with AODV [?]. Complementarily, Suh *et al.* combine unsupervised *clustering* with deep RL to reduce route-update overhead by 30 % when nodes move chaotically [?].

Finally, **citizen participation** via *mobile crowdsensing* has expanded MANET capabilities. A systematic survey by Cicek and Kantarci reviews 25 studies in which smartphones themselves collect environmental data, geolocation, or images to enhance the situational awareness of emergency agencies; the review confirms that, when combined with intelligent aggregation techniques, these volunteer networks can provide reliable information with acceptable energy consumption [?]. Thus, disaster-related applications range from mission-critical links for first responders to participatory citizen-reporting platforms, all supported by the self-configurable and increasingly intelligent nature of ad-hoc networks.

2.2.5 Challenges in Highly Dynamic Environments

The main distinctive feature of *ad-hoc* networks in disaster scenarios is their **highly volatile topology**: links are created and torn down within milliseconds due to irregular three-dimensional movement, node loss, and unpredictable obstacles. In UAV scenarios, for instance, vertical mobility increases the likelihood of partitioning and multiplies the number of hops required to maintain connectivity [?]. This volatility triggers bursts of stale routes and produces reconvergence delays that degrade critical QoS metrics such as end-to-end delay.

Another challenge is the **signalling overhead**. Recent comparative studies show that when traffic increases from 5 to 20 pkt/s, the overhead of reactive protocols surges by up to 65% in 100-node networks, owing to route-discovery requests and periodic maintenance messages [?]. The problem is exacerbated by variable node densities—such as sudden concentrations of rescuers or emergency vehicles—which saturate the channel and cause collisions.

Interference and congestion or jamming attacks add further uncertainty. A study by Abuzainab *et al.* demonstrates that interference losses can reduce throughput by as much as 40% even under moderate mobility levels [?]. Effective countermeasures therefore require adaptive mechanisms capable of identifying free channels and rebuilding routes in real time.

Density–time variability also challenges machine learning: models must generalise from sparse topologies to dense clusters without incurring a state-space explosion. The survey by Zheng *et al.* reports that RL-based MAC algorithms struggle to scale once the number of nodes exceeds one hundred [?]. To mitigate this issue, hybrid strategies combine unsupervised *clustering* with deep RL, cutting table-update overhead by 30% [?]. Likewise, in high-speed vehicular environments, Upadhyay *et al.* employ an enhanced DRL scheme that lowers average latency and increases PDR even at densities above 120 vehicles/km [?].

Finally, **energy constraints and device heterogeneity**—from low-power sensors to battery-rich UAVs—complicate the selection of stable and efficient routes. Without dynamic load balancing and power control, critical nodes deplete their batteries, causing

catastrophic failures in the communication mesh. These challenges underscore the need for cognitive protocols that integrate mobility, interference, and energy metrics to keep the network operational under extreme conditions.

2.2.6 Communication and Routing Protocols

Routing protocols in *ad-hoc* networks are traditionally grouped into **proactive**, **reactive**, and **hybrid** categories. Proactive schemes—such as DSDV and OLSR/OLSRv2—maintain full routing tables and offer low discovery latency, which makes them a common baseline in disaster-network testbeds and field trials [?]. Their main drawback is the constant control traffic they generate: comparative studies under high-mobility simulations show that, at 20 pkt/s, proactive overhead can exceed that of reactive protocols by about 35 % [?]. Hardware-centric enhancements such as HMC-AODV cut end-to-end delay by up to 25 % through FPGA parallelism and homogeneous clustering [?], while adaptive variants of OLSR that reduce “HELLO” frequency have been proposed to make the protocol more suitable for highly dynamic emergency scenarios [?].

In disaster contexts, **reactive protocols** (AODV, DSR, DYMO) are preferred for their low initial overhead. However, their performance degrades under priority traffic and collision conditions. To address this, Ozen and Ozen propose **PA-AODV**, which assigns priority levels to emergency packets and raises the *Packet Delivery Ratio* by 12 % in ns-3 scenarios with 150 nodes [?]. Jin *et al.* introduce *DRLRR*, a deep-reinforcement-learning-based scheme that improves PDR by 18 % and shortens latency by 22 % compared with AODV in post-earthquake environments [?].

The rise of **artificial intelligence** has spurred the development of adaptive routes that react to mobility in real time. Notable examples include (i) **PARRoT**, which combines reinforcement learning with UAV trajectory prediction and lowers average latency by 17 % [?]; (ii) **QLR-FANET**, where Q-learning and rate control stabilise highly mobile aerial links [?]; and (iii) the resilient protocol by Suh *et al.*, which applies unsupervised *clustering* before deep RL and cuts route-update overhead by 30 % in chaotic networks [?]. More recently, Dikmen *et al.* show that a distributed DRL scheme with dense rewards preserves connectivity in topologies of up to 250 nodes while sacrificing less than 8 % of throughput compared with the theoretical optimum [?].

Hybrid and multi-domain-aware protocols integrate interference, energy, and congestion metrics. Upadhyay *et al.* introduce IDRL-VANET, which optimises routes and controls collisions in vehicular networks, achieving latency reductions at densities of ≥ 120 vehicles/km [?]. These trends point toward cognitive, software-defined architectures in which ML algorithms operate alongside adaptive MAC/PHY layers to sustain the resilience required for emergency missions.

2.2.7 IEEE 802.11e EDCA and QoS Provisioning

The IEEE 802.11e standard—released as an amendment in 2005—introduced the **Enhanced Distributed Channel Access** (EDCA) mechanism to provide quality of service in WLANs and, by extension, in *ad-hoc* networks [?]. EDCA extends the legacy Distributed Coordination Function (DCF) by adding four *Access Categories* (ACs): voice, video, best effort, and background. Each category has its own contention parameters—*Arbitration Inter-Frame Space* (AIFS), minimum/maximum *Contention Window*, and *Transmit Opportunity* (TXOP)—that prioritise delay-sensitive flows without the need for central coordination. In disaster contexts, this differentiation is essential to ensure that rescue-team voice frames or UAV aerial images receive priority over less critical traffic.



Figure 2.6: Priority control scheme of EDCA [?].

Early analyses showed that simply configuring EDCA parameters statically may be insufficient under mixed loads and high mobility. Ni *et al.* demonstrated that dynamically adjusting the contention-window size according to traffic type significantly improves latency and video/voice throughput without harming lower categories [?]. Later, Inan *et al.* developed a cycle-time-based model that accurately computes delay and capacity for each category even under saturation, providing a theoretical foundation for EDCA auto-tuning algorithms [?].

In emergency *ad-hoc* networks, EDCA is often combined with deep reinforcement learning to reconfigure AIFS and CW according to node density and interference levels, thus merging the standard's intrinsic prioritisation with AI-driven adaptability. This synergy makes it possible to maintain smooth voice service and stable video transmission even when the topology and radio environment change unpredictably.

2.3 Mobility in Emergency Contexts

The **human factor** shapes every emergency-communications architecture, since the effectiveness of an *ad-hoc* network depends on the cognitive, affective, and mobility patterns of the affected population. Recent studies show that the *intention to evacuate* is dominated by perceived efficacy of the action and the subjective probability of harm; together, these factors explain most of the variance in evacuation decisions during a hurricane [?]. This finding supports the *Protective Action Decision Model* and underscores the importance of messages that emphasise the benefits of relocating in the face of risk.

Once the decision turns into action, collective dynamics exhibit *self-organising* behaviour. Multi-exit simulations with agents trained in urban drills reveal that familiarity with specific exits can unbalance pedestrian flows, causing bottlenecks and lowering the overall evacuation efficiency [?]. These congestions—combined with the simultaneous voice and data traffic generated by rescue teams—put additional stress on the network; therefore, mobility models for MANETs must account for crowd effects and partial route blockages.

The advent of large-scale *anonymous geolocation* databases has made it possible to quantify these phenomena in real time. An analysis of more than 40 million mobile-device records during Hurricane Ian showed that departure peaks occur within two hours of an official order, with spatially heterogeneous evacuation rates—reaching 68 % in coastal counties but only 23 % in inland areas [?]. Such disparities call for dynamic aerial-coverage schemes—e.g., FANETs—that can redistribute capacity where user density multiplies.

Finally, *citizen participation* adds another behavioural layer: more than 70 % of damage reports in recent earthquakes originate from social networks or *mobile crowdsensing*. A systematic review identifies 25 studies in which smartphones themselves contribute environmental and geo-tagged data, showing that intelligent aggregation can deliver reliable information at acceptable energy costs [?]. Integrating these collaborative data streams into the routing logic strengthens network resilience and enhances the situational awareness of authorities.

2.3.1 Importance of Realistic Mobility Models

Rigorous evaluation of *ad-hoc* networks requires node movements to accurately reproduce the paths and physical constraints of a disaster environment. Using random patterns—e.g., the *Random Waypoint* model—can **overestimate performance**: a comparative study showed that such models “do not reflect obstacles or the logic of rescue teams, leading to unreliable results” [?]. Indeed, the review concludes that the *Disaster Area Mobility Model* (DA-MM) yields more plausible trajectories by incorporating walls, debris, and command posts, thereby improving the external validity of the tested protocols.

Tools such as **BonnMotion 3** make it easy to generate these realistic scenarios and export them directly to simulators (ns-2, ns-3, QualNet, COOJA, etc.) [?]. In addition to the classic models, BonnMotion provides disaster-specific variants—*Disaster Area*, *Random Street*, and *Manhattan Grid*—that capture topographic constraints and assign node speeds according to role (victim, rescuer, UAV).

In vehicular environments, road models integrated with ns-3 allow driving kinematics (IDM, lane changes) to be coupled with network events. Arbabi and Weigle show that “mobility–message” synchronisation reduces the deviation between simulation and real data, forming the basis for analysing emergency routes on highways [?]. Similar results are obtained by fusing SUMO with ns-3 for urban contexts, where evacuee density varies dramatically by area.

In summary, **selecting a realistic mobility model**:

- 1) prevents over- or under-estimating QoS metrics,
- 2) aligns the network topology with the population’s cognitive and logistical patterns, and
- 3) enables AI algorithms to be trained on traces that closely resemble real-world behaviour.

The use of DA-MM, mass-geolocation-based traces, and tight mobility–network coupling has therefore become an indispensable quality criterion for research in emergency communications.

2.3.2 Mobility Models in Simulated Networks

Simulation results depend critically on the pattern by which nodes move; therefore, mobility models are now broadly divided into three families [?]:

(i) *generic synthetic models*—e.g., *Random Waypoint*, *Gauss–Markov*—which offer mathematical simplicity but ignore obstacles; (ii) *scenario-specific models*, which incorporate geography, roles, and tactical constraints (for example, collapsed walls or exclusion zones); and (iii) *empirical traces* obtained from geolocation data or traffic simulators such as SUMO.

Models for disaster areas. The classic *Disaster Area Mobility Model* (DA-MM) has recently been refined by Affandi *et al.*, who propose the **DRT model** for search-and-rescue operations; its zoned version reduces overhead by 83 % and improves PDR by 5 % compared with the original DA-MM in NS-2 [?]. These advances confirm the need to capture rescuers’ operational logic—grid patrols, mobile command posts—to avoid overestimating network capacity.

Traffic–network integration in vehicular environments. For VANETs, fidelity hinges on coupling vehicle micro-kinematics with network events. The **NDN4IVC** framework links SUMO and ns-3 through TraCI and synchronises every simulation step, enabling the evaluation of V-NDN protocols over realistic mobility traces [?]. This bidirectional approach avoids the inconsistencies typical of “two-phase” methods and facilitates AI training on coherent data.

The current trend is clear: **move away from purely random patterns** and employ models that combine physical constraints, operational roles, and empirical validation. This ensures that the resulting QoS metrics—delay, *throughput*, and PDR—reliably represent network behaviour when deployed in an actual disaster.

2.3.3 Mobility Integration in Network Performance Evaluation

To realistically evaluate a MANET, it is essential to couple the spatial dynamics of the nodes with the network engine. The most common workflow relies on the **BonnMotion** + **ns-3** combination, which can reproduce complex movements and measure quality-of-service metrics on the same time scale.

a) **Trace generation and conversion.** BonnMotion makes it possible to create synthetic or disaster-specific scenarios—such as the *Disaster Area Mobility Model*—and export them to `.ns2` or `.xml` format. Using the command `bm-converter -m ns3 earthquake` the trace is converted into a file readable by `ns-3`, which is then loaded with `MobilityHelper::Install()` [?]. This operation ensures that every coordinate of the trajectory is replicated exactly in the network simulation.

c) **Metric instrumentation.** The `FlowMonitor` and `AsciiTraceHelper` utilities capture end-to-end delay, PDR, and *throughput*. In addition, mobility-derived variables—such as *link lifetime* and *pause time*—can be dumped to CSV files to feed machine-learning models or for subsequent statistical analysis [?]. Running at least ten replicas with different random seeds allows 95 % confidence intervals to be estimated.

By integrating BonnMotion with `ns-3`, one obtains a coherent environment in which the topology evolves as it would in a real disaster and the network metrics respond accordingly, thereby avoiding the overly optimistic conclusions that arise when mobility and network layers are treated separately.

2.4 Network Simulation in Critical Environments

2.4.1 Simulation Tools: NS-3 and BonnMotion

Pre-deployment validation of *ad-hoc* networks in emergency scenarios relies almost exclusively on **packet-level simulation**. Among the benchmark platforms, the discrete-event simulator `ns-3` and the mobility generator BonnMotion stand out; together, they can reproduce the entire protocol stack and node movements with millisecond granularity.

NS-3 Simulator

NS-3 (Network Simulator 3) is a discrete-event network simulator designed primarily for research and education. It is an open-source project started in 2006 and distributed under the GNU GPLv2 license [?]. NS-3 is implemented mainly in C++, although Python bindings are also available, providing additional flexibility for developing simulations. **Figure 2.7** shows a typical NS-3 setup for a wireless-sensor-network scenario.

As the successor to **ns-2**, **ns-3** features a C++/Python core that models the PHY, MAC, IP, and transport layers, with native support for Wi-Fi, LTE, and 5G [?]. Its *tracing* architecture enables detailed capture of metrics such as PDR, end-to-end delay, and *throughput*, which are indispensable for disaster-recovery protocols. Extensions like the **energy** framework allow battery-consumption estimation in sensors or UAVs, while the **dce/ns-3** module can run real binaries (e.g., *quagga*, *iperf*), bringing performance even closer to operational conditions [?]. Likewise, the **TraCI** interface integrates SUMO for vehicle-traffic co-simulation, preventing mobility–network desynchronisation [?].

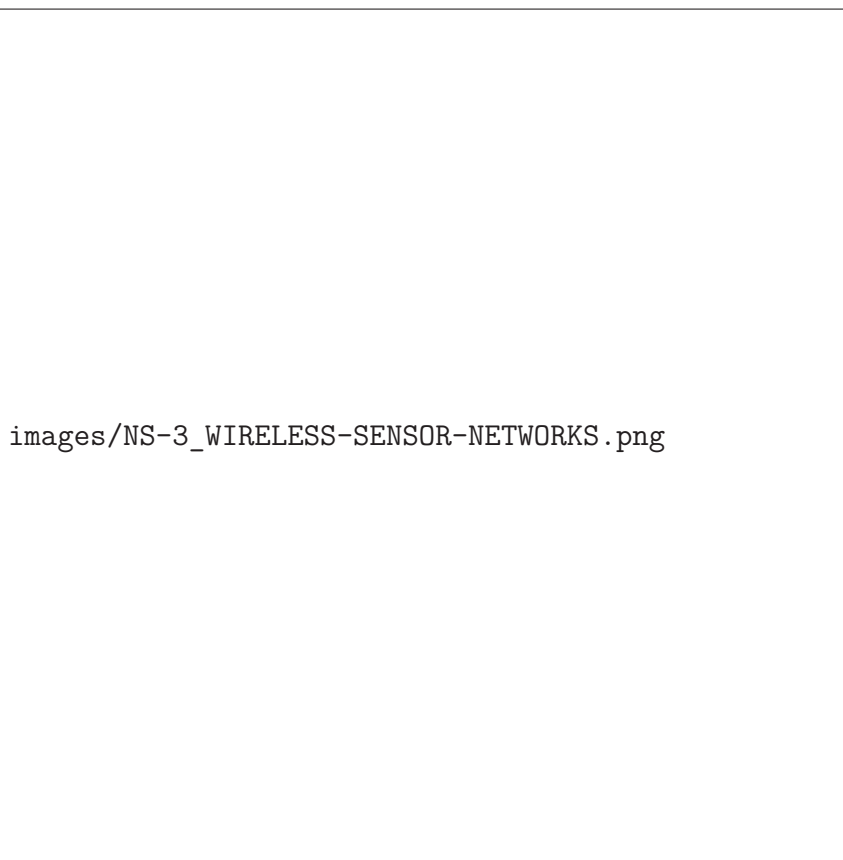


Figure 2.7: NS-3 simulation example [?].

This simulator provides a robust platform for modelling communication networks, including network protocols, devices, topologies, mobility models, and propagation models. Its modular architecture allows researchers to define and customise complex network scenarios, making it possible to analyse protocol behaviour under diverse conditions without the need for physical hardware [?].

Its most salient features include:

- Support for both wired and wireless networks
- Detailed protocol-stack models (TCP/IP, Wi-Fi, LTE, 5G, etc.)
- Integration with external tools such as Wireshark and NetAnim
- Ability to simulate large-scale networks with high temporal precision

Thanks to its accuracy, flexibility, and active support community, NS-3 has become an essential tool in academia and research for studying communication networks.

BonnMotion

BonnMotion is a Java-based software tool for generating and analysing mobility scenarios. It is widely used in research on mobile ad-hoc networks (MANETs), wireless sensor networks (WSNs), and vehicular ad-hoc networks (VANETs). Its main function is to create movement traces that can be exported to different network simulators, including **ns-2**, **ns-3**, **GloMoSim/QualNet**, **COOJA**, **MiXiM**, and **The ONE** [?].

BonnMotion produces mobility traces based on both synthetic and scenario-specific models; version 3 adds the *Disaster Area Mobility Model* (DA-MM)—illustrated in **Figure 2.8**—and exports directly to **.ns2** and **.xml** formats compatible with **ns-3** [?]. Notable profiles include *Random Street*, *Manhattan Grid*, and *Disaster Area*, which incorporate obstacles, exclusion zones, and roles (victim, rescuer, UAV). A study by Mahiddin *et al.* shows that using DA-MM traces reduces PDR over-estimation by 18 % compared with *Random Waypoint* in AODV tests [?].



Figure 2.8: BonnMotion disaster scenario diagram [?].

The tool was initially developed by the Communication Systems group at the University of Bonn, Germany, and has since been maintained and extended by other institutions such as Osnabrück University and the Colorado School of Mines. BonnMotion supports a wide variety of mobility models, including:

- **Random Waypoint**
- **Random Walk**
- **Gauss-Markov**
- **Manhattan Grid**
- **Reference Point Group Mobility (RPGM)**
- **Disaster Area**
- **Random Street**

- **Steady-State Random Waypoint**, entre otros[?].

Thanks to its flexibility and compatibility with multiple simulators, BonnMotion has become an essential tool for researchers seeking to evaluate network-protocol behaviour under realistic mobility conditions.

Mobility–network synergy. The typical workflow involves generating a trace with BonnMotion, converting it to **ns-3** format using the **bm-converter** script, and feeding it to **MobilityHelper**. This allows simulator nodes to reproduce the exact trajectories while the upper layers record QoS degradation whenever links are severed. Such synergy has been crucial for training *deep RL* models that anticipate partitions and decide when to deploy reinforcement UAVs [?]. Consequently, the **ns-3+BonnMotion** pair has become the de facto standard for AI-based research in emergency communications.

2.4.2 Limitations of Simulated Environments

Although the **ns-3+BonnMotion** combination is the most widely used testbed for *ad-hoc* networks, several factors can distort the conclusions if they are not interpreted with caution. First, channel and MAC models rely on statistical simplifications; phenomena such as coherent multipath, co-channel interference, and hardware impairments are only approximated, which can lead to discrepancies between simulated values and field measurements [?]. In addition, as the number of nodes grows, computational load rises quickly and often forces researchers to lower the temporal resolution or limit the generated traffic, directly affecting delay and PDR.

Mobility and traffic fidelity are equally decisive: synthetic models or CBR sources do not always capture the data spikes and human behaviour typical of mass evacuations, resulting in overly optimistic QoS estimates [?]. Moreover, porting the same scenario to other simulators (e.g., Veins/OMNeT++) can yield divergent results because of internal differences in timing and propagation, complicating cross-study comparison [?].

To mitigate these limitations, key parameters should be validated against real measurements, multiple replicas with independent seeds should be executed, and simulations should be complemented with emulation or small field pilots whenever feasible. In this

way, the obtained results better approximate the network behaviour expected once it is deployed in a real disaster scenario.

2.5 Artificial Intelligence in Wireless Networks

Artificial intelligence (AI) encompasses techniques that allow systems to learn from data, reason with models, and make autonomous decisions [?]. Within this umbrella, three main paradigms are recognised: *supervised learning* (predicts outputs from labelled inputs), *unsupervised learning* (discovers patterns without labels), and *reinforcement learning* (optimises actions through rewards) [?]. These categories share a common workflow: data collection, feature preparation and selection, model training, and validation.

In the field of **wireless networks**, AI operates across multiple layers of the OSI stack:

- **Physical layer:** channel estimation and interference suppression via neural networks that replace traditional filters [?].
- **MAC layer:** dynamic adjustment of contention windows, traffic prioritisation, and spectrum allocation in Wi-Fi or LTE schemes.
- **Network layer:** route selection, load balancing, and early congestion detection in both core and *ad-hoc* networks.
- **Management and orchestration:** power optimisation, *network slicing*, and self-organisation (SON) in 4G/5G infrastructures.
- **Security:** traffic classification, attack detection, and automatic anomaly response.

2.5.1 Techniques: Supervised, Unsupervised, and Reinforcement Learning

AI algorithms applied to *ad-hoc* networks can be broadly classified into three paradigms—*supervised*, *unsupervised*, and *reinforcement* learning—each addressing a different class of problems within the protocol stack.

Supervised learning. This paradigm is employed when pairs $\langle \text{input}, \text{label} \rangle$ describing the network state are available. Almeida *et al.* use a CNN to estimate *throughput* and delay from ns-3 traces, achieving an RMSE 35 % lower than that of linear regressors [?].

Iqbal *et al.* train an *XGBoost* ensemble on 120,000 MANET-traffic samples and detect link failures with an F1 score of 0.97, enabling early route reconfiguration [?]. Such supervised solutions are well suited for **QoS prediction**, **anomaly classification**, and **link-state estimation**.

Unsupervised learning. When labels are scarce, *clustering* and *dimensionality-reduction* methods uncover latent structure. Suh *et al.* apply *k-means* to cluster nodes before training their DRL agent, reducing signalling overhead by 30 % [?]. Du *et al.* propose a **variational autoencoder** that generates traffic embeddings and detects topological anomalies, improving the detection rate by 12 % relative to PCA [?]. Unsupervised learning is therefore useful for **forming hierarchical clusters**, **inferring roles**, and **identifying interference patterns** without prior knowledge.

Reinforcement learning (RL). RL is well suited to the dynamic, partially observable environments typical of disasters. Protocols such as **DRLRR** use a DQN to select the next route and raise PDR by 18 % in post-earthquake urban scenarios [?]. **PARRoT** combines trajectory information with RL, reducing average latency in UAV swarms [?]. More recent **multi-agent** schemes—such as IDRL-VANET [?] and the proposal by Dikmen *et al.* [?—coordinate routing and power decisions among neighbouring nodes, preserving connectivity with only an 8 % throughput penalty when scaling to 250 nodes.

Taken together, **supervised learning** delivers accurate predictions when labelled data are available; **unsupervised learning** uncovers hidden structure and segments the network to reduce complexity; and **reinforcement learning** closes the control loop, adapting routes and resources in real time. The combination of these paradigms—often in hybrid architectures (e.g., GNN + RL or clustering + DQN)—represents the current frontier in optimising *ad-hoc* networks for disaster scenarios.

2.5.2 Ethical and Practical Considerations

Applying AI to *ad-hoc* networks for disaster response offers operational benefits but also raises moral and practical demands that must be addressed before deployment.

Responsibility and safety. European guidelines for trustworthy AI state that systems must be robust and include clear *accountability* mechanisms [?]. When a deep-RL agent makes routing decisions, any failure can paralyse emergency communications; therefore, *circuit breakers* are recommended to hand control back to proven protocols (AODV/OLSR) whenever the intelligence behaves anomalously [?].

Privacy and data protection. Mass aggregation of geolocation traces—crucial for training mobility models—poses re-identification risks. Anonymisation mechanisms based on *differential privacy* reduce spatial filtering but raise density-estimation error by up to 9 % [?]. Techniques such as federated learning or local differential privacy are advisable to balance utility and confidentiality.

Fairness and bias. Models trained on unbalanced data tend to prioritise high-density urban areas, relegating rural communities; this creates QoS disparities in UAV deployment or bandwidth allocation. Fairness audits show that an unbalanced DRL algorithm delivered 15 % less PDR in peripheral zones [?]. Reward re-weighting or *oversampling* techniques can mitigate this effect.

Transparency and explainability. Policies learned by DNNs are often opaque. *Explainable-AI* methods (saliency maps, SHAP) reveal which metrics (link lifetime, local density) drive the agent’s actions and facilitate regulatory certification [?]. Mission-critical environments therefore call for an interpretability threshold that enables human review.

Robustness against attacks. RL agents can be vulnerable to *poisoning* or adversarial jamming attacks that alter rewards and deviate routes [?]. Adversarial training and GNN-based anomaly detection improve resilience, albeit with added computational overhead.

Computational and energy cost. On-line inference of large models can drain portable nodes. DOro *et al.* show that a 1.2-M-parameter DQN agent increases power consumption by 18 % on ARM Cortex-A53 nodes [?]. Pruned networks or hardware accelerators (e.g., NPUs) can lower this penalty without sacrificing decision quality.

In summary, AI-based optimisation must be accompanied by ethical safeguards—transparency, fairness, privacy—and by a practical analysis of resources and threats. Only then will it be both acceptable and sustainable for critical communications in disaster scenarios.

2.6 Quality of Service and Performance Evaluation

2.6.1 Metrics: Delay, Packet Delivery, and *Throughput*

Communication quality in *ad-hoc* networks is assessed primarily with three key indicators:

(a) End-to-end delay (D_{e2e}). This is the average time elapsed between a packet's generation at the sender and its correct reception at the destination:

$$D_{\text{e2e}} = \frac{1}{N_{\text{rx}}} \sum_{i=1}^{N_{\text{rx}}} (t_i^{\text{rx}} - t_i^{\text{tx}}),$$

where N_{rx} is the number of packets received and t_i^{tx} and t_i^{rx} are the transmission and reception instants, respectively. Values of $D_{\text{e2e}} \leq 150$ ms are deemed acceptable for push-to-talk voice in rescue operations [?]. ns-3 studies show that congestion and hop count raise the delay almost linearly ($R^2 = 0.92$) beyond the third retransmission [?].

(b) Packet Delivery Ratio (PDR). This metric measures data-transport reliability:

$$\text{PDR} = \frac{N_{\text{rx}}}{N_{\text{tx}}} \times 100\%,$$

where N_{tx} is the total number of packets sent. Recent DRL protocols maintain $\text{PDR} \geq 90\%$ up to densities of 150 nodes, whereas AODV drops below 80 % in the same scenario [?]. The ITU recommends $\text{PDR} \geq 95\%$ for critical-telemetry applications [?].

(c) *Throughput* ($\mathcal{T}$). Amount of useful bits received per unit time:

$$\mathcal{T} = \frac{\sum_{i=1}^{N_{\text{rx}}} L_i}{T_{\text{sim}}} \quad [\text{bit/s}],$$

where L_i is the length (bits) of packet i and T_{sim} is the simulation duration. Razouqi *et al.* report that *throughput* drops by 37 % when the HELLO interval is shortened from 1 s to

0.25 s because of higher control overhead [?]. Streaming 720p video at 30 fps requires at least 2 Mb/s per flow—a crucial reference when sizing inspection FANETs [?].

Complementary metrics. *Jitter* (delay variation), *goodput* (useful bits over total bits), and energy consumed per delivered bit are increasingly used in AI evaluations, as they capture the temporal stability and efficiency of a protocol [?]. In disaster networks, normalising these indicators facilitates cross-study comparison and helps identify configurations that prioritise quality of service over raw throughput.

2.6.2 Network Evaluation in Simulated Scenarios

The credibility of experiments carried out with `ns-3+BonnMotion` hinges on an **evaluation methodology** that covers statistical design, scenario diversity, and sensitivity analysis.

1. Statistical design and replicas. The `ns-3` statistics framework recommends running at least 10 replicas with independent seeds and reporting *95 % confidence intervals (CIs)* [?]. Petrov *et al.* confirm that a sample size of $n \geq 10$ keeps CI error below 5 % for PDR and delay [?]. Pullen and Buchner warn that increasing the integration step to speed up large simulations (>500 nodes) can underestimate delay by around 20 %, reinforcing the need to balance accuracy and computational cost [?].

2. Parameter and scenario coverage. To avoid biased conclusions, a *factorial design* is recommended, combining:

- **Node density:** low (50), medium (150), high (300).
- **Mobility patterns:** DA-MM with mild/medium/severe intensities [?].
- **Traffic load:** CBR 64 kb/s, push-to-talk voice 16 kb/s, video 2 Mb/s.
- **Channel models:** Nakagami ($m = 1$), LogDistance, ObstacleShadowing [?].

Razouqi *et al.* show that conclusions changed from “AODV superior” to “OLSR superior” when density increased from 100 to 200 nodes—even though the metrics and simulator were identical [?]. Hence, exhaustive parameter exploration is essential.

3. Sensitivity analysis and external validity. Lai *et al.* propose varying each parameter by $\pm 10\%$ and measuring the elasticity of the metrics; they identify transmit power and the HELLO interval as explaining 70 % of the variance in PDR [?]. Moreover, comparing simulation results with *field benchmarks* or DCE emulation reduces the simulator–reality gap [?].

2.6.3 Impact of Mobility and Environment on QoS

Quality of service (QoS) in *ad-hoc* networks varies markedly with **speed**, **node density**, and the environment’s **physical characteristics**. The most relevant findings are summarised below.

Speed and movement pattern. Increasing the average speed from 2 m/s to 15 m/s causes a near-linear drop in *Packet Delivery Ratio* (PDR) from 95 % to 74 % in AODV, owing to more frequent link breaks [?]. Kumar *et al.* observed that a smoothed 3-D Gauss–Markov model raises PDR by 7 % over the classical Gauss–Markov model by mitigating abrupt direction changes in UAV swarms [?]. These results underscore the importance of realistic mobility models (DA-MM, ASSGM-3D) to avoid overestimating QoS.

Density and congestion. Razouqi *et al.* demonstrated that as node density increases from 50 to 300, end-to-end latency grows almost quadratically—from 38 ms to 235 ms—owing to control-channel saturation [?]. Unsupervised clustering strategies, such as the one proposed by Suh *et al.*, cut overhead by 30 % and stabilise latency at 180 ms even in dense topologies [?].

Physical environment and obstacles. Carpenter and Sichitiu’s *ObstacleShadowing* model shows that concrete buildings add 14–22 dB of attenuation, reducing *throughput* by 35 % in urban scenarios relative to free space [?]. Lai *et al.* found that the combination of high mobility and shadow zones explains 70 % of the variance in PDR, underscoring the need for joint mobility–channel co-simulation [?].

Weather and environmental conditions. Ribeiro *et al.* evaluated fire-response networks and found that dense smoke propagation increases packet-loss rates by up to 12 %,

highlighting the need for real-time power adjustment and channel selection [?]. In coastal settings, marine reflectance can improve low-altitude coverage, raising PDR by about 5 % in maritime-rescue FANETs [?].

Studies show that *QoS is tightly coupled* to mobility dynamics and the physical environment. Simulation campaigns should therefore:

1. Use realistic mobility models and cover a range of speeds and densities,
2. Include obstacle losses and climatic variability, and
3. Apply AI techniques (clustering, DRL) that adapt to such changes.

Only under these conditions will delay, PDR, and throughput metrics reflect the performance expected when the network is deployed in a genuine disaster scenario.

Chapter 3

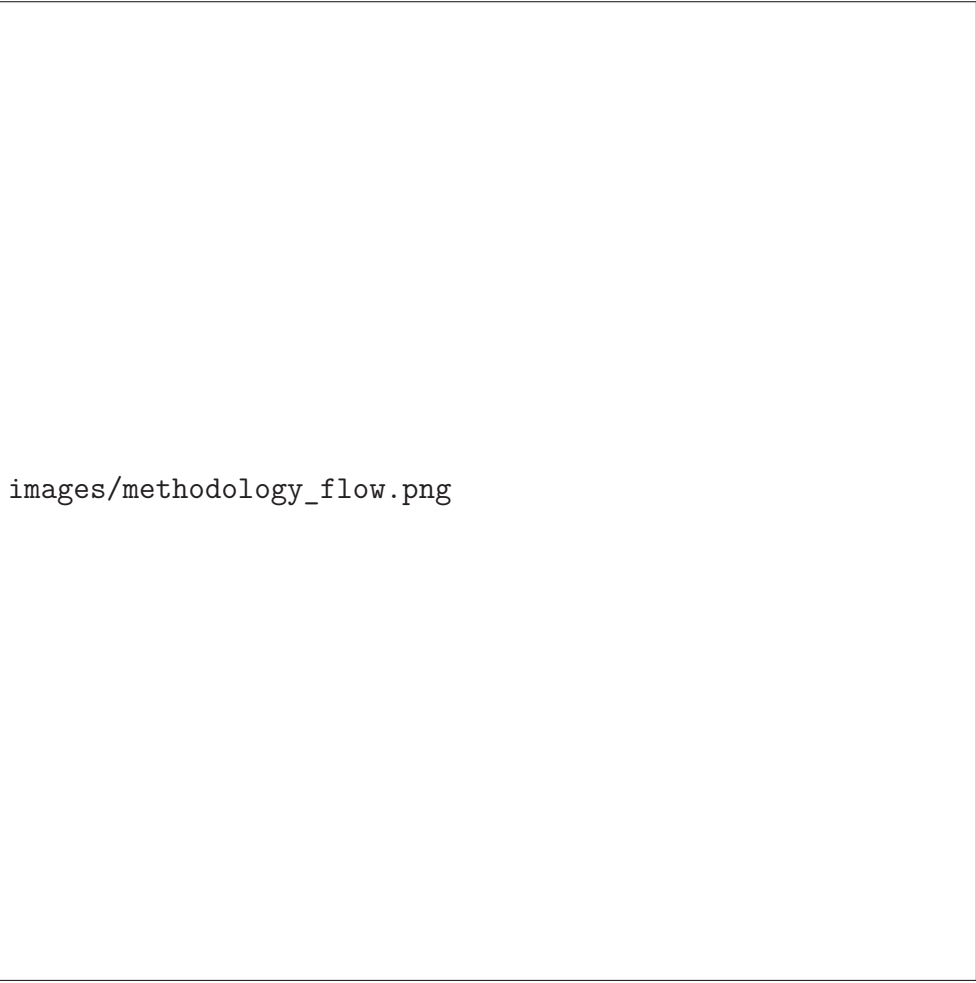
Methodology

Figure 3.1 illustrates the overall workflow for implementing artificial intelligence in disaster scenarios. To simulate human mobility in post-disaster environments, we used the **BonnMotion** simulator, which generates realistic movement patterns of individuals in dynamic and chaotic conditions. For modeling communication between individuals, we employed the **ns-3** network simulator, configuring it to represent ad hoc wireless networks in infrastructure-less settings. During the simulation process, we collected relevant communication parameters, including the *channel utilization factor* and the *queue packet size*, which serve as the primary input features for our AI model.

In the data processing phase, the collected information was labeled to construct a supervised learning dataset. We trained a machine learning model using this dataset, incorporating *hyperparameter optimization techniques* to improve model performance. The trained model was then evaluated using standard machine learning metrics. Finally, the resulting model was exported and integrated back into the **ns-3** simulation environment, allowing us to test its performance under new disaster-driven communication scenarios. Additional details about each component of the methodology are presented in the following subsections.

3.1 Typical Disaster-Area Scenario

The **Disaster Area Mobility Model** (DA-MM)—included in BonnMotion 3—was designed to *accurately reproduce* the movements of victims and first responders after a catastrophic event [?]. Unlike random-waypoint models, DA-MM partitions the scene into *func-*



images/methodology_flow.png

Figure 3.1: Overview of the proposed methodology for integrating AI into disaster scenarios.

tional zones: debris fields, command posts, evacuation routes, and safe areas. Each zone enforces specific speed limits, turning angles, and congestion probabilities, so that node trajectories capture both physical obstacles and the tactical logic of search-and-rescue operations.

Key parameters.

- **-n**: number of nodes per role (*victim*, *rescuer*, *uav*).
- **-d**: simulation duration (s).
- **-x/-y**: dimensions of the disaster zone (m).
- **-vR** and **-vV**: speed distributions for rescuers and victims, respectively.
- **-obst**: polygons that represent collapsed buildings or blocked streets.

The scenario depicted in **Figure 3.2** is presented as an introductory example and is based on a large-scale disaster drill. The exercise took place in May 2005 in Cologne, Germany, as part of the preparations for World Youth Day 2005 and the FIFA World Cup 2006. The underlying assumption was that a catastrophe in an event hall had injured more than 250 people [?].



Figure 3.2: Introductory disaster area scenario [?].

The different zones involved in this scenario are listed in **Table 3.1**. The overall area spans approximately 350×200 m. The incident location is the event hall; directly in front of it lies the zone where patients wait for treatment. Injured individuals are transported to four casualty-clearing stations. In addition, a technical operational command centre and an ambulance parking point are provided.

Acronym.	Area
IL	Incident location
PWFTA	Patients waiting for treatment area
CCS	Casualties clearing stations
APP	Ambulance parking point
TEL	Technical operational command

Table 3.1: Areas in disaster scenario.

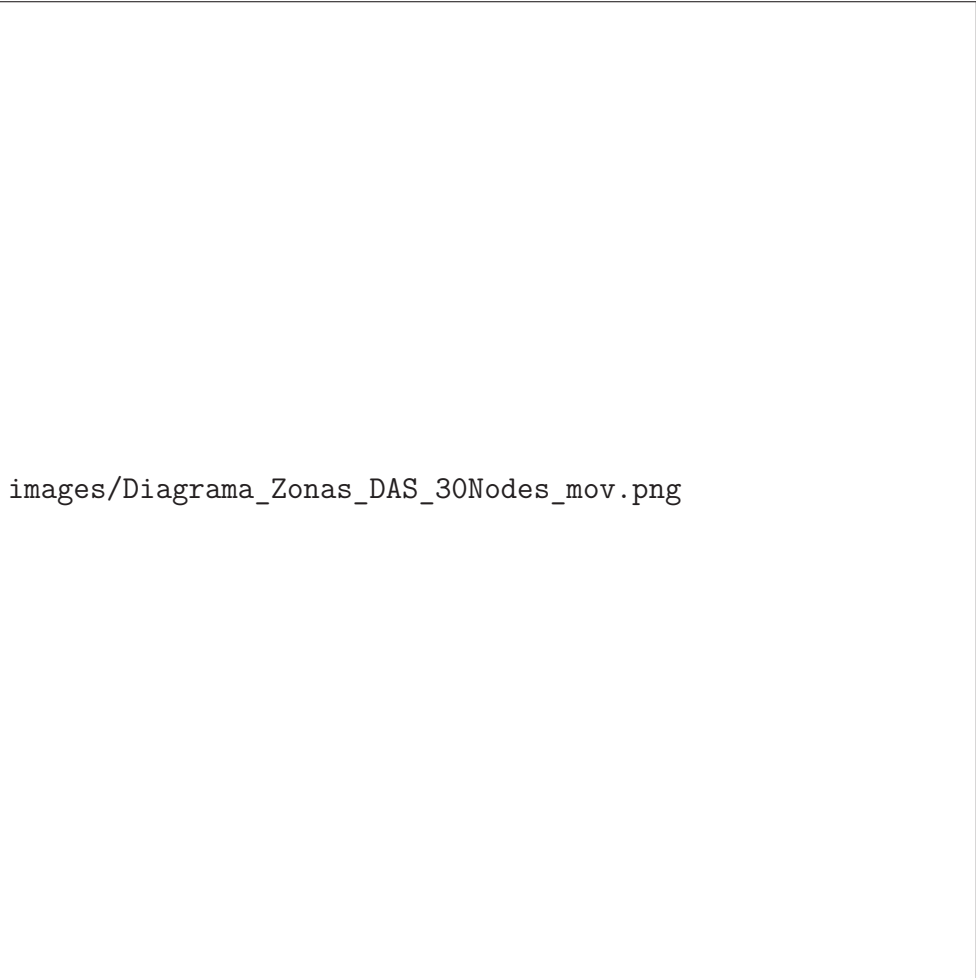
3.2 Parameters and configuration of the disaster scenario in BonnMotion.

Figure 3.3 depicts a 350×200 m disaster scenario inspired by the drills conducted in Cologne (2005) for World Youth Day 2005 and the 2006 FIFA World Cup. The tactical zones—*Incident Location* (IL), *Patients Waiting For Treatment Area* (PWFTA), *Casualties Clearing Station* (CCS), *Ambulance Parking Point* (APP), and the centralised *Technical Operational Command* (TEL)—mirror the real flow of casualties and resources during the first hours after the event, as summarised in Table 3.1. This layout enables differentiated mobility patterns: chaotic movements in IL, ambulance routes between CCS and APP, and coordination traffic concentrated at TEL.

In this study, the grey rectangle labelled *Obstacle* is retained solely as a spatial landmark and does not affect the propagation layer. Explicitly modelling diffraction and obstacle-specific shadowing would add computational complexity without providing decisive information for the QoS analysis; therefore, the evaluation focuses on the spatial dynamics of the nodes and the traffic load under realistic mobility conditions.

Trajectories within the disaster scenario. Figure 3.4 displays the trajectories generated by BonnMotion for the 30 nodes in the *DisasterArea* scenario. Each coloured rectangle represents a specific tactical zone—*Incident Location* (IL, blue), *Patients Waiting For Treatment Area* (PWFTA, green), *Casualties Clearing Station* (CCS, beige), *Ambulance Parking Point* (APP, purple), and the *Technical Operational Command* (TEL, red)—while the grey arrows indicate the nodes’ random movements over the 300 s simulation.

- **Intra-zone dynamics.** Within each area, the nodes follow a uniform random pattern, mirroring the chaotic mobility of victims and rescuers who move without a predefined

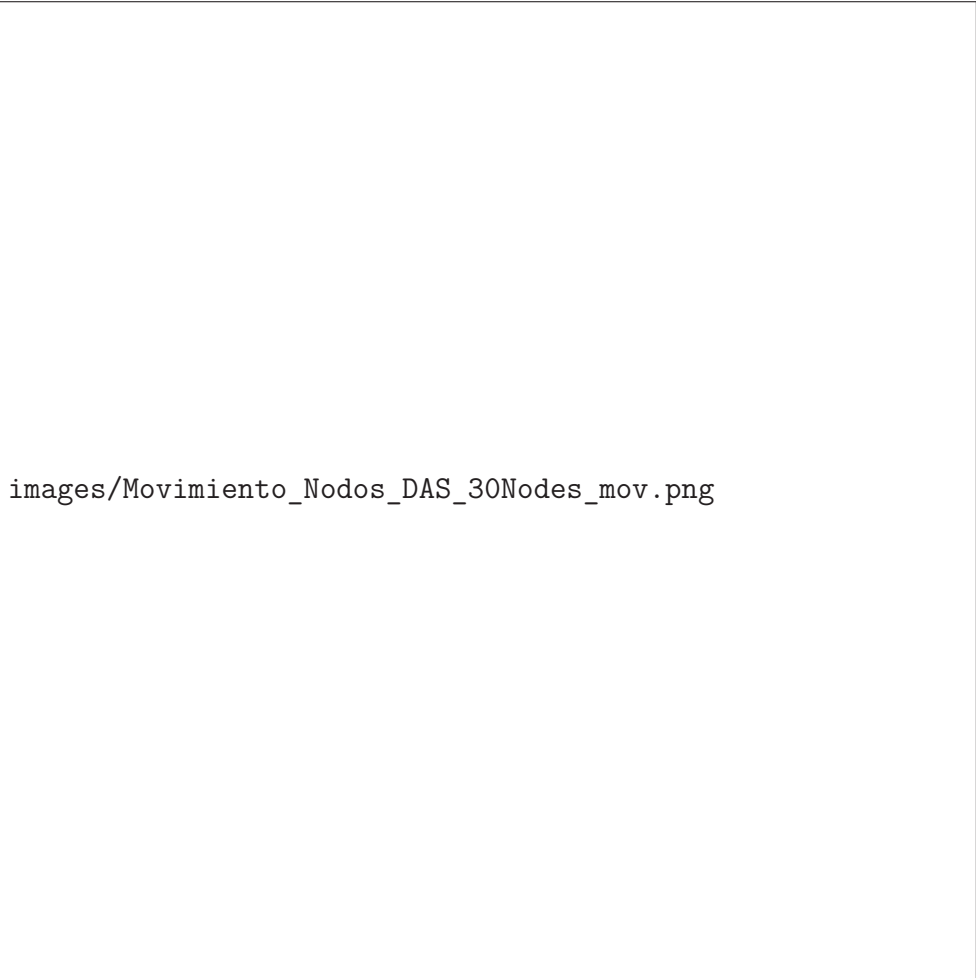


images/Diagrama_Zonas_DAS_30Nodes_mov.png

Figure 3.3: Proposed disaster scenario.

route yet remain confined to their operational perimeter.

- **Inter-zone flows.** The lines connecting different zones emulate real transfers: from IL to PWFTA and CCS (evacuation of casualties), from CCS to APP (ambulance boarding), and from all zones to TEL (data uploads and resource requests). These transitions are crucial for triggering link breaks and assessing the resilience of routing protocols.
- **Spatio-temporal variability.** The overlay of trajectories reveals changing densities—local congestion at the TEL (command centre) versus dispersion at the APP (peripheral area). Such variability forces network algorithms to adapt to both dense clusters and long, low-connectivity links.
- **Obstacle.** The grey rectangle labelled “Obstacle” is kept as a geographic landmark—representing debris or a collapsed wall—but is not enabled in the propagation model,



images/Movimiento_Nodos_DAS_30Nodes_mov.png

Figure 3.4: Random movement of the nodes.

so the analysis focuses on pure mobility effects at the network layer.

Taken together, this controlled random pattern reproduces the inherent unpredictability of a disaster and yields a highly dynamic topology: short-lived links within zones and longer (up to 200 m) inter-area connections. Such dynamism is indispensable for testing *Machine-Learning* algorithms capable of anticipating partitions, adjusting power, or re-routing critical traffic in real time.

Parameter configuration. As shown in Table 3.2, the “DisasterArea” scenario is configured for 30 nodes in a 350×200 m area and a duration of five minutes, representing the critical initial phase of a mass-casualty incident. Group changes are disabled ($\$groupchange = 0$) to reflect the autonomy of each terminal, and the random seed is fixed ($\$seed = 23$) to ensure experimental reproducibility. These settings balance opera-

tional realism and computational cost, providing a solid foundation for assessing routing protocols and AI algorithms under mobility conditions that closely resemble reality.

Variable	Value	Description
<code>\$model</code>	<code>DisasterArea</code>	Mobility model chosen in BonnMotion.
<code>\$scenario</code>	<code>DAS_30Nodes</code>	Trace prefix for version control and tracking.
<code>\$nodes</code>	30	Number of nodes: operational staff plus sensors.
<code>\$x</code>	350	Length of the area (m), taken from the Cologne drill.
<code>\$y</code>	200	Width of the area (m).
<code>\$groupchange</code>	0	No group change: nodes act autonomously.
<code>\$groupsize</code>	1	Each node forms its own group (individual).
<code>\$dist</code>	3 m	Minimum intra-group distance to avoid overlaps.
<code>\$mindist</code>	3 m	Absolute minimum distance between any pair of nodes.
<code>\$circlevertices</code>	140	Polygon resolution for circular areas.
<code>\$factor</code>	1	Speed-scale factor (no modification).
<code>\$duration</code>	300 s	Simulation window (first 5 min of the emergency).
<code>\$skip</code>	1 s	Temporal sampling step in the trace.
<code>\$seed</code>	23	Random seed for reproducibility.
<code>\$maxpause</code>	0 s	Maximum pause: nodes always in active movement.

Table 3.2: Global parameters used to generate the mobility trace in BonnMotion.

Figure 3.5 presents the mobility-trace parameters as configured in BonnMotion.

Table 3.3 lists the rectangular boundaries (in metres) and the operational purpose of each tactical block within the *DisasterArea* model. The IL, PWFTA, and CCS areas are aligned along the x -axis to reflect the horizontal evacuation flow from the incident site to the casualty-clearing station, whereas the TEL is placed in a central, elevated location, maximising its betweenness in the communication mesh. The APP is positioned at the far eastern edge to enforce long outbound links. This spatial arrangement allows differentiated mobility patterns and traffic loads to be assigned, thereby increasing the realism of the simulations relative to the actual flows in a rescue operation.

Figure 3.6 shows the coordinates and operational roles of the tactical zones in the scenario as configured in BonnMotion.

Command-line construction and execution. Figure 3.7 illustrates how all the required flags are concatenated into the `$params` variable so that BonnMotion can generate the `DAS_30Nodes` trace with the *DisasterArea* model. The workflow follows the classical *configure* $\rightarrow$ *build* $\rightarrow$ *run* pattern:



Figure 3.5: Parameters used to generate the mobility trace in BonnMotion.

Acronym	X Range	Y Range	Operational Role
IL1	75–125	25–75	Incident Location (epicentre of the event)
PWFTA1	150–200	25–75	Patients Waiting for Treatment Area (triage)
CCS1	225–275	25–75	Casualties Clearing Station (stabilisation)
TEL	150–200	100–150	TechnicalOperationalCommand (command centre)
APP1	320–345	5–50	Ambulance Parking Point (victim loading)
OBST1	25–100	150–200	Obstacle (debris / collapsed wall)

Table 3.3: Coordinates and operational functions of the tactical zones in the scenario.

- `-f $scenario DisasterArea` simultaneously sets the output file *prefix* and selects the *mobility model*.
- `-n $nodes -x $x -y $y` instantiates 30 nodes within a 350×200 m rectangle.
- `-p $maxpause` sets the maximum pause to 0 s, ensuring continuous movement.
- `-g $groupsize -g $circlevertices -r $dist -q $mindist` defines group micro- dy-



images/DisasterAreaConfiguration2.png

Figure 3.6: Coordinates of the tactical zones in the scenario.

namics and the minimum inter-node distance.

- `-d $duration -e $naxareas -l $skip -j $factor` control the simulation length (300 s), the number of tactical areas (5), and the temporal resolution (1 s).
- The five occurrences of `-b ...` describe the rectangles IL, PWFTA, CCS, TEL, and APP, while `-o $OBST1` adds a reference obstacle.
- The flag `-K` instructs BonnMotion to produce a `.stat` file with global statistics.
- Finally, `-R $seed` fixes the random seed (23), ensuring reproducibility.

The call `system("$bm $params");` executes the `bm` binary with the full command string, producing three essential outputs: `DAS_30Nodes_mov.mov` (a position trace with 1 s granularity), `DAS_30Nodes_mov.ns2 / .xml` (directly importable into `ns-3`), and `DAS_30Nodes_mov.stat` (aggregate metrics). This procedure—explicit parameterisation combined with automatic command construction—ensures that the experiments can be reproduced

on any Linux environment running BonnMotion 3, thereby promoting transparency and facilitating direct comparison with related work.



Figure 3.7: Execution parameters used in the BonnMotion command.

3.3 Data Collection

In the data collection phase, we used the **BonnMotion** tool to simulate the mobility of individuals in a disaster scenario. This tool allows the generation of realistic human movement patterns, which are essential to model behavior in chaotic environments. The main parameters configured in BonnMotion for this study are summarized in Table 3.4.

To simulate communication between individuals, we used the **ns-3** simulator, configuring a wireless ad hoc network without fixed infrastructure. The network layer employed the **OLSR** (Optimized Link State Routing) protocol, which is suitable for highly dy-

namic topologies typical in disaster situations. At the transport layer, the **UDP** protocol was used. Packet sizes followed an **exponential distribution** with an average of approximately 200 bytes, reflecting the variability in message lengths common in emergency communications. Additionally, we used the **IEEE 802.11a** standard to model the physical layer of the wireless network. The main simulation parameters configured in ns-3 are shown in Table 3.5.

To improve the diversity and robustness of the dataset, multiple simulation runs were performed with different random seeds. This approach enabled the collection of a wide range of communication scenarios and mobility patterns. The simulation outputs were stored and organized to compute key performance metrics, including *channel utilization* and *queue packet size*, which serve as input features for the machine learning model.

Parameter	Value / Description
Mobility Model	[Disaster Mobility Model]
Number of Nodes	[30]
Simulation Duration	[300 seconds]
Area Dimensions	[350 m $\times$ 200 m]
Speed Range	[0.5 to 1.5 m/s]
Pause Time	[0 s]

Table 3.4: Key Parameters Used in BonnMotion Simulation

Parameter	Value / Description
Network Protocol	OLSR (Optimized Link State Routing)
Transport Protocol	UDP
Packet Size Distribution	Exponential (mean: 200 bytes)
PHY/MAC Standard	IEEE 802.11a
Channel Bandwidth	[20 MHz]
Simulation Time	[300 seconds]
Number of Nodes	[30]
Seed Variations	Multiple seeds for robustness
Application Traffic	[Variable bit rate]
Queue Type	[FIFO]

Table 3.5: NS-3 Simulation Configuration Parameters

3.4 Data Processing

After the simulation phase, a large volume of raw data was collected from multiple network runs. These data represent low-level information related to node behavior, packet transmissions, queue dynamics, and channel access events. To prepare the dataset for machine learning, it was necessary to apply specific feature extraction techniques to derive meaningful indicators of network status.

For our study, we focused on two key features that characterize the state of the network at any given time:

- **Channel Utilization Factor** : This metric estimates the level of congestion in the wireless medium. It is calculated by monitoring the percentage of time the channel is sensed as busy versus idle. To smooth temporal fluctuations and capture recent behavior trends, we applied an *Exponential Weighted Moving Average (EWMA)* over the raw channel activity measurements.
- **Queue Packet Count** : This metric reflects the number of packets queued for transmission at each node. A high queue occupancy may indicate potential congestion, which could lead to packet drops or increased delays. The queue management policy used was the default *FIFO* (First-In-First-Out) policy provided by the **ns-3** simulator.

These two features were extracted at regular time intervals throughout each simulation and paired with the corresponding network outcomes. In particular, each data instance was labeled according to whether a packet successfully reached its destination or not, under the prevailing network conditions. This labeled dataset forms the basis for training our machine learning model, enabling it to learn how network metrics influence packet delivery success in disaster-driven ad hoc environments.

3.5 Model Training

The classification task in this study is binary in nature: given the current network conditions, the model must predict whether a packet will be successfully delivered to its

destination or not. Various supervised learning algorithms could be applied to this type of problem; however, we chose to employ an ensemble-based approach using the **CatBoost** classifier.

CatBoost, developed by Yandex, is a high-performance gradient boosting algorithm based on decision trees. It is specifically designed to handle categorical data efficiently and reduce overfitting by implementing advanced techniques such as ordered boosting. One of its main advantages lies in its efficient handling of decision splits through symmetric trees and histogram-based processing, which makes it especially suitable for deployment on resource-constrained devices.

To train the CatBoost model, we focused on tuning the most critical hyperparameters: the **tree depth**, set to values between 6 and 10, and the **number of trees**, which varied up to 5000 estimators. In order to determine the optimal combination of these hyperparameters, we performed an exhaustive **Grid Search**. This method was selected over more lightweight alternatives due to the availability of sufficient computational resources, allowing for a comprehensive evaluation of the hyperparameter space.

The final trained model was validated using a hold-out portion of the dataset and evaluated using standard classification metrics, as discussed in the next section.

3.6 Model Evaluation

The performance of the trained CatBoost model was evaluated using standard machine learning metrics for binary classification, including **accuracy**, **precision**, and the **F1-score**. These metrics provide a comprehensive view of the model's ability to distinguish between successful and unsuccessful packet deliveries under varying network conditions.

Accuracy measures the proportion of correct predictions out of all predictions made, offering a general indication of overall model performance. *Precision* quantifies the ratio of true positive predictions to all positive predictions made, reflecting the model's reliability when it predicts that a packet will be successfully delivered. The *F1-score* is the harmonic mean of precision and recall, providing a balanced assessment that is especially useful in the presence of class imbalance. These metrics were computed on a held-out test set, allowing for an unbiased evaluation of the model's generalization capabilities. The results

obtained indicate that the model achieves high predictive performance, making it suitable for real-time deployment in simulated disaster communication environments.

Chapter 4

Results and Discussion

4.1 Numerical Results

After training the artificial intelligence model using the custom-built dataset and validating its performance through machine learning metrics, the final model was exported and integrated into the **ns-3** simulator. The aim was to evaluate its effectiveness in a simulated disaster scenario by comparing network performance in two different settings: one without Quality of Service (QoS) mechanisms and another enhanced with the AI-driven decision-making process.

To quantify the benefits of the proposed approach, we evaluated both scenarios using key network performance indicators. These include the **Packet Delivery Ratio (PDR)**, which represents the percentage of packets successfully delivered to their destinations; the **Throughput**, which measures the effective data rate achieved by the network; and the **End-to-End Delay**, which quantifies the average time taken for packets to traverse the network from source to destination.

The detailed analysis and comparison of these metrics are presented in the following subsections.

4.2 Packet Delivery Ratio (PDR)

Figure 4.1 presents the evaluation of the Packet Delivery Ratio (PDR) across four applications in a disaster scenario. Human mobility patterns were generated using the BonnMotion simulator, and wireless ad hoc communication was modeled using ns-3. The experiment

compares three QoS strategies: a baseline scenario without congestion control (**NO_CC**), a traditional approach using EDCA (**EDCA_CC**), and an AI-driven method (**ML_CC**) based on a CatBoost model trained on relevant network features.

As depicted in the figure, the AI-based solution (**ML_CC**) consistently outperforms the other two approaches in terms of PDR. While the EDCA mechanism provides moderate improvements over the **NO_CC** scenario in certain applications, its performance degrades significantly under specific conditions, particularly in App 4. In contrast, the **ML_CC** approach demonstrates robustness and adaptability by maintaining high PDR across all applications.

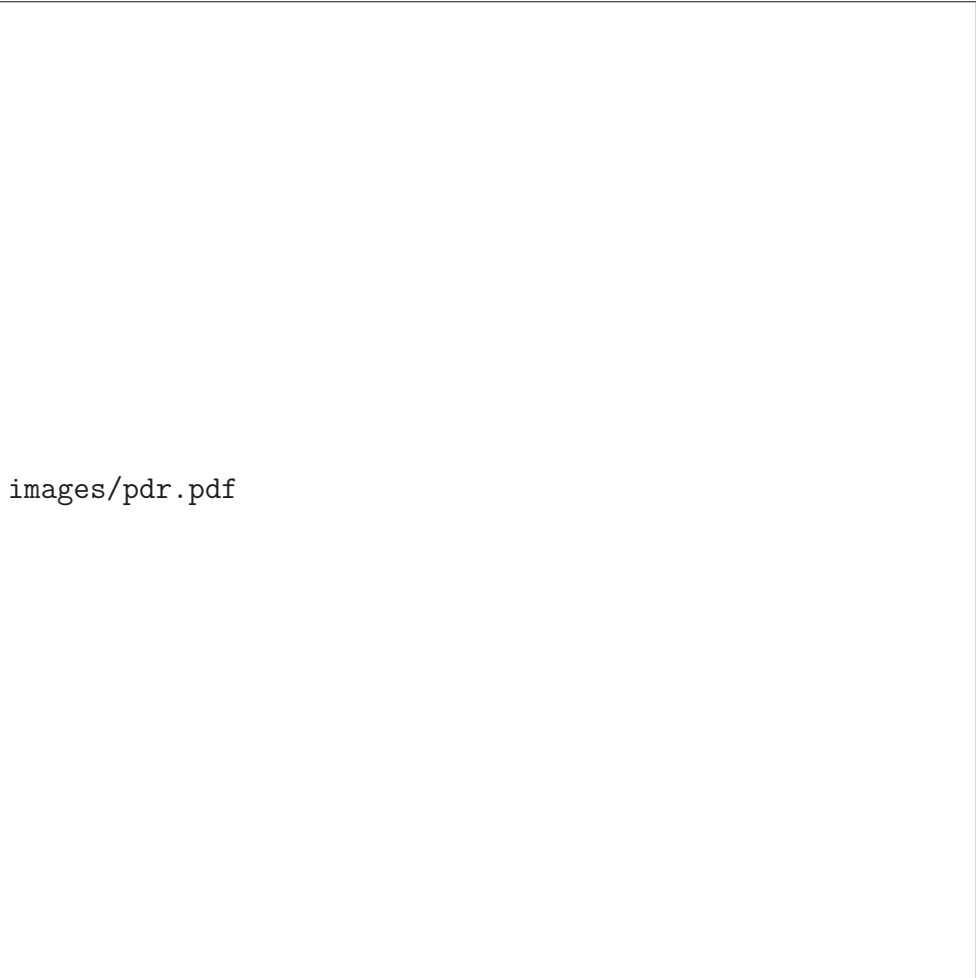
These results suggest that the AI-enhanced strategy is better equipped to handle the dynamic and congested conditions typical of disaster environments. By leveraging real-time features such as channel occupancy and queue length, the model is able to make informed decisions about packet transmission, thereby improving reliability and overall QoS performance.

4.3 Throughput

Throughput, defined as the rate of successfully received data at the destination (in kbps), is a critical metric for assessing the efficiency of communication in disaster-prone ad hoc networks. These environments are highly dynamic, with frequent topology changes, variable signal conditions, and intense channel contention—making high throughput difficult to sustain.

Figure 4.2 illustrates the throughput performance of the three QoS strategies evaluated: the baseline without congestion control (**NO_CC**), the EDCA-based mechanism (**EDCA_CC**), and the machine learning-based control (**ML_CC**) using the **CatBoost** algorithm.

Overall, the **ML_CC** approach achieves comparable or higher throughput than the other scenarios, particularly in applications where traditional methods struggle. For example, in **App 4**, **ML_CC** significantly outperforms both **EDCA_CC** and **NO_CC**, highlighting its ability to adapt under high congestion or degraded channel conditions. While **EDCA_CC** shows slight improvements over the baseline in some applications (Apps 1



images/pdr.pdf

Figure 4.1: Comparison of Packet Delivery Ratio (PDR) under three QoS scenarios in a disaster environment: no congestion control (NO_CC), EDCA-based control (EDCA_CC), and machine learning-based control (ML_CC).

and 2), it suffers in others due to its static nature and lack of predictive capabilities.

These findings reinforce the value of intelligent, data-driven control mechanisms that can dynamically adjust to real-time network states. The increased throughput seen with **ML_CC** aligns with the improvements observed in **Packet Delivery Ratio (PDR)**, and further demonstrates the effectiveness of machine learning in reducing retransmissions, optimizing packet scheduling, and improving overall communication efficiency in critical scenarios.



images/throughput.pdf

Figure 4.2: Comparison of received throughput across three QoS strategies (**NO_CC**, **EDCA_CC**, and **ML_CC**) in a simulated disaster scenario.

4.4 End-to-End Delay

End-to-End Delay refers to the average time a data packet takes to travel from the source to the destination node. This metric encompasses all types of delays, including those caused by buffering, retransmissions, routing decisions, and medium access contention. In highly dynamic and unpredictable disaster scenarios, reducing delay is essential for enabling prompt communication, especially in applications that involve safety-critical or emergency data.

Figure 4.3 displays the end-to-end delay observed under three different Quality of Service (QoS) strategies: no congestion control (**NO_CC**), EDCA-based control (**EDCA_CC**), and machine learning-based control (**ML_CC**) using the **CatBoost** algorithm.

The results show a significant reduction in delay for the **ML_CC** scenario across

all applications. While the EDCA approach achieves moderate improvements over the baseline, it fails to maintain low delay under more congested or variable conditions (e.g., **App 4**). In contrast, the **ML_CC** strategy exhibits consistent low latency, indicating its effectiveness in avoiding transmission under unfavorable conditions such as high queue occupancy or busy channels.

These findings demonstrate that AI-driven strategies can dynamically adapt to the network state, minimize queuing and processing delays, and ultimately enhance responsiveness in wireless ad hoc environments affected by disaster-induced mobility and interference.



images/end_to_end_delay.pdf

Figure 4.3: Comparison of end-to-end delay under three QoS strategies (**NO_CC**, **EDCA_CC**, and **ML_CC**) in a disaster scenario.

Chapter 5

Conclusions

- This work presented an intelligent approach for improving communication in ad hoc wireless networks deployed in disaster scenarios. By simulating realistic mobility patterns using **BonnMotion** and modeling infrastructure-less communication through **ns-3**, we constructed a custom dataset from scratch. This dataset captured key performance indicators of the network, particularly the *channel utilization factor* and *queue packet size*, which were used to train a **CatBoost**-based machine learning model for binary classification.
- The trained model was validated using machine learning metrics such as accuracy, precision, and F1-score, and subsequently exported back into the simulation environment to support real-time decision-making. Comparative experiments showed that the AI-enhanced communication strategy significantly outperforms the baseline (without QoS) in terms of **Packet Delivery Ratio (PDR)**, **Throughput**, and **End-to-End Delay**.
- These results demonstrate the potential of integrating AI into network protocol layers for enabling adaptive, context-aware communication strategies in highly dynamic environments. As future work, we plan to extend our model to support multi-class decision-making, incorporate additional features such as link stability or energy consumption, and evaluate real-time implementations on embedded or low-cost hardware platforms.

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